# 14 Mendel and the Gene Idea

## KEY CONCEPTS



## Study Tip

**Solving genetics problems:** In Figure 14.5, you'll learn how to predict the offspring of a genetic cross with a Punnett square, like the example shown below. For crosses shown in other figures, cover up the Punnett square and draw your own. Also, check out the Tips for Genetics Problems at the end of the chapter.

| | Sperm *P* | Sperm *p* |
|---|---|---|
| Eggs *P* | *PP* | *Pp* |
| Eggs *p* | *Pp* | *pp* |

## Go to Mastering Biology

**For Students** (in eText and Study Area)
- Get Ready for Chapter 14
- Figure 14.4 Walkthrough: Alleles, Alternative Versions of a Gene
- Animation: Cross of One Character in "MendAliens"

**For Instructors to Assign** (in Item Library)
- Tutorial: Mendel's Law of Independent Assortment
- Tutorial: Pedigree Analysis: Galactosemia

**Figure 14.1** Flower color was one of the many characteristics of pea plants studied by Gregor Mendel. By breeding pea plants over many generations and carefully counting the types of offspring, Mendel developed a theory of inheritance that is the basis of modern genetics.

### How are traits, such as the purple or white color of flowers, transmitted from parents to offspring?

Alternative versions of a gene (alleles) account for different traits.